Budget-Aware Controller Blending: Post-Hoc Damage-Budget Enforcement for Pre-Trained RL Policies

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Abstract

We consider the common deployment scenario in which a *pre-trained, unconstrained* RL policy faces a new cumulative operational constraint after deployment: a damage budget per year, a duty-cycle limit, a spending cap. Retraining the policy to satisfy the constraint is the standard answer, but on the standard Safety Gymnasium benchmark even Lagrangian finetune starting from the deployed actor collapses reward to ≈ 0 within 50,000 steps and fails to recover after 500,000—the dual penalty destroys goal-seeking before budget satisfaction emerges. We propose instead a closed-form schedule that *blends two existing controllers at inference*: the deployed performance policy π_{perf} and a certified safety controller π_{safe} (e.g., a factory wind-tracking yaw controller, an artificial-potential-field navigator). The blend weight $\sigma(u) \in [0, 1]$ is a monotone function of the *urgency ratio* $u = \rho/\tau$ (remaining budget fraction over remaining time fraction). On pace ($u \geq 1$): run the performance policy. Budget deficit ($u \to 0$): shift to the safety controller. A single scalar η controls sharpness; a single policy serves any budget d instantly. On wind-farm yaw control ($\kappa \approx 0.72$), the schedule retains 99.3–101.8% of unconstrained power while satisfying a 15-step neg-yaw budget across 4 training seeds. On Safety Gymnasium ($\kappa \approx 0.02$), the APF-blend variant retains 45–71% of unconstrained reward while paying down the budget, where Lagrangian finetune retains $< 2\%$ even with 500k extra steps. A gradient-correction variant of the same schedule works when the action-to-cost coupling strength $\kappa = \mathbb{E}\|\nabla_a Q_c\|/\mathbb{E}\|\nabla_s Q_c\|$ is large; when it is small, the explicit-π_{safe} blend is required.

1 Introduction

Scenario. A wind-farm operator deploys an RL controller that maximizes power via coordinated yaw wake steering; it was trained unconstrained and works well. Six months later, a bearing inspection tightens the maintenance schedule: turbine 2's yaw may be negative for at most 15 hours per year. Retraining the RL controller is expensive and slow; swapping controllers is disruptive; *doing nothing* violates the constraint. The operator already has a certified wake-steering fallback (e.g., WakeAdapt [Siemens Gamesa, 2023], a calibrated yaw-setpoint table). What they need is a way to *pace between* the aggressive RL policy and the certified safety controller so that the cumulative constraint is satisfied, without retraining either.

This scenario is pervasive. Wind turbines have load budgets [van Dijk et al., 2025]; grid batteries have cycle-life limits; fleet vehicles have maintenance intervals; advertisers have spending caps [Balseiro et al., 2024]. Unlike an instantaneous safety constraint, a *lifetime budget* requires choosing *when* to spend limited capacity in a risky regime. A fixed scalarization $\alpha P - (1 - \alpha)D$ is memoryless: it cannot tighten in response to a realized-damage trajectory exceeding the planned path.

Why retraining is not the default answer. We find experimentally (Figure 2, §5.2) that finetuning the pretrained Safety-Gymnasium actor with SAC-Lagrangian destroys the reward within 50,000 extra steps and

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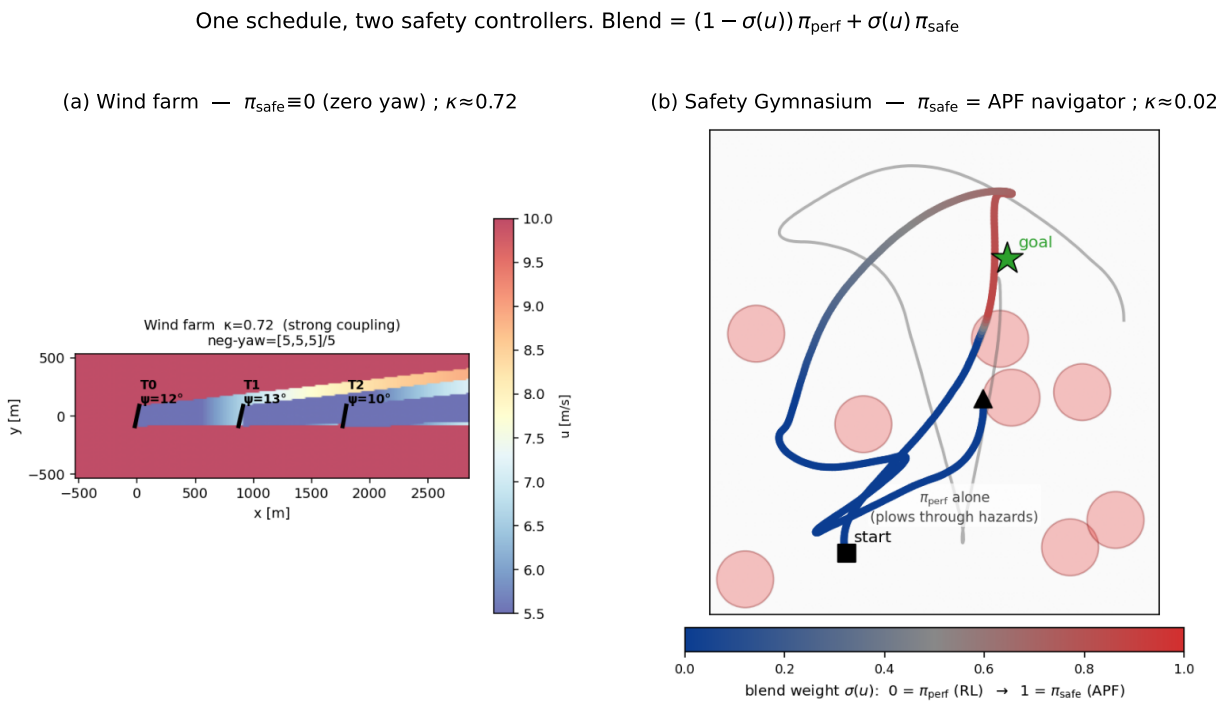

Figure 1: Two domains, two safety controllers, one schedule. (a) Wind farm: π_{safe} is a zero-yaw (no wake-steering) setpoint; the Jensen wake field shows the farm under a budget-satisfying blend ($d = 5$, neg-yaw $[5, 5, 5]$). (b) Safety Gymnasium: π_{safe} is an artificial-potential-field navigator. The trajectory is color-coded by blend weight $\sigma(u)$: blue where the RL actor drives ($\sigma \to 0$), orange where the APF safety controller drives ($\sigma \to 1$). The $\sigma(u)$ track (middle) rises sharply when the cumulative cost (bottom) approaches the budget, pulling the agent off the goal-seeking path into hazard-avoiding APF mode until the urgency ratio recovers.

stays collapsed at 500,000: the Lagrangian dual drives the policy toward a "stand still" safe minimum before useful budget-aware behaviour emerges. The problem is not that CMDP is impossible—it is that under operator-relevant compute budgets, CMDP retraining replaces a useful policy with a useless-but-safe one.

Our proposal: urgency-ratio controller blending. Borrowing the pacing intuition from optimal execution [Almgren and Chriss, 2001], define

$$u_t = \frac{\rho_t}{\tau_t}, \qquad \rho_t = \frac{d - C_t}{d}, \qquad \tau_t = \frac{T - t}{T}, \tag{1}$$

and blend two pre-existing controllers at inference:

$$a_t = (1 - \sigma(u_t))\,\pi_{\text{perf}}(s_t) + \sigma(u_t)\,\pi_{\text{safe}}(s_t). \tag{2}$$

$\sigma : \mathbb{R}_{\geq 0} \to [0, 1]$ is monotone decreasing in u (Section 3). On pace $\Rightarrow$ performance; deficit $\Rightarrow$ safety.

Contributions. (1) A *closed-form urgency-ratio schedule* $\sigma(u)$ with a single sharpness parameter, derived from budget-balanced allocation under an idealized Boltzmann response (§3; Prop. 1). (2) *Two-controller blending* that treats an existing safety controller as first-class primitive rather than retraining the policy (§4). (3) A *gradient-correction variant* (§4.2) in which π_{safe} is implicitly defined by a step of $-\nabla_a Q_c$; this variant works precisely when the coupling strength $\kappa = \mathbb{E}\|\nabla_a Q_c\| / \mathbb{E}\|\nabla_s Q_c\|$ is large (Table 1). (4) *Empirical validation* on wind-farm yaw control ($\kappa \approx 0.72$) and Safety Gymnasium ($\kappa \approx 0.02$): under the *same* schedule with *different* safety controllers chosen by κ, both domains satisfy the budget. A Lagrangian-finetune baseline, starting from the same pretrained actor, is dominated at every compute checkpoint we measured through 500k steps (§5).

Deployment claim. The method drops into systems that already have an RL performance policy, a certified safety controller, and a damage estimator—none of which are retrained. The safety layer is already certified; the urgency schedule decides when to fall back to it.

2 Background

Constrained MDPs. A CMDP [Altman, 1999] adds cost-return constraints to the MDP objective: $\max_\pi \mathbb{E}_\pi[\sum_t r_t]$ subject to $\mathbb{E}_\pi[\sum_t c_t] \leq d$ (standard notation [Ray et al., 2019, Achiam et al., 2017]). Lagrangian methods [Stooke et al., 2020] learn a dual μ *during training*; changing d requires a new policy. SAUTE [Sootla et al., 2022] augments state with ρ_t (the same variable we use) but still retrains per budget.

Post-hoc composition. Energy composition [Urain et al., 2023, Gladstone et al., 2025], classifier guidance [Zhang et al., 2025], CBFs [Xiao et al., 2025], and input-convex corrections [Durkin et al., 2025] modify pre-trained policies at inference to satisfy *instantaneous* constraints. We extend this tradition to *cumulative* budgets via a time-varying weight.

Multi-objective scalarization. A fixed $\alpha P - (1-\alpha)D$ trade-off is memoryless. Our schedule turns α into a time-varying $\alpha(u_t) = (1-\sigma(u_t))\,\alpha_{\text{perf}} + \sigma(u_t)\,\alpha_{\text{safe}}$ by pacing against realized damage.

3 Urgency-Ratio Schedule

Proposition 1 (Pacing motivation)**.** *Under an idealized Boltzmann response $P(\text{risky} \mid \lambda) = P_0 \exp(-\alpha\lambda)$, the penalty weight that maintains budget-rate balance takes the multiplicative form $w^*(u) = 1/u$ relative to the TWAP baseline ($u{=}1$).*

Proof sketch. Balance $(T-t)P(\text{risky} \mid \lambda) = d - C_t$ with the Boltzmann substitution gives $\lambda = \frac{1}{\alpha} \ln \frac{(T-t)P_0}{d-C_t}$. The multiplicative weight relative to TWAP reduces to $\tau/\rho = 1/u$. Full derivation in Appendix A.1. □

This is a *motivating* pacing argument, not a derivation of the actual schedule we deploy. The Boltzmann-response model is a strong assumption; real RL policies do not respond linearly in log-probability to scalar penalties. What we use from Prop. 1 is the *state-variable identification*: the urgency ratio u is the natural argument and the schedule should be monotone-decreasing in u. The specific functional form below is a parametric choice with one tunable scalar.

Practical schedule. Since $1/u \to \infty$ as $u \to 0$, we use the parametric family $w(u) = \exp(\eta(1/u - 1))$ with $\eta \geq 0$, clamped at $\lambda_{\max} = 10^6$. It agrees with $u^{-\eta}$ within 5% for $u \in [0.7, 1.3]$ (Appendix Fig. 3); outside that band the schedule is a deliberate heuristic chosen for smooth bounded transitions.

Blend weight. For Eq. 2 we map $w(u)$ to a bounded $\sigma \in [0,1]$:

$$\sigma(u) = \begin{cases} 0 & u \geq 1, \\ 1 - \exp(-\eta(1/u - 1)) & u < 1. \end{cases} \tag{3}$$

On pace ($u \geq 1$): pure performance policy. Deficit ($u < 1$): sigmoid-like ramp toward pure safety controller, with sharpness η. A hard-wall backstop drives $\sigma \to 1$ for $u < 0.05$.

4 Method

The method has three independent components: **(i)** the schedule $\sigma(u)$ (how much safety); **(ii)** the performance policy π_{perf} (what we deploy on pace); **(iii)** the safety controller π_{safe} (what we revert to on deficit). *Any* π_{safe} composes with *any* π_{perf} under the *same* schedule.

Algorithm 1 Urgency-Ratio Controller Blending

Require: Performance policy π_{perf}, safety controller π_{safe}, budget d, horizon T, sharpness η

$C_0 \leftarrow 0$
for $t = 0, 1, \ldots, T-1$ **do**
  $u_t \leftarrow \frac{(d-C_t)/d}{(T-t)/T}$
  $\sigma_t \leftarrow 0$ if $u_t \geq 1$ else $1 - \exp(-\eta(1/u_t - 1))$
  $a_t \leftarrow (1-\sigma_t)\,\pi_{\text{perf}}(s_t) + \sigma_t\,\pi_{\text{safe}}(s_t)$
  Execute a_t; observe c_t; $C_{t+1} \leftarrow C_t + c_t$
end for

4.1 Explicit Safety-Controller Blend (primary formulation)

$$a_t = (1-\sigma(u_t))\,\pi_{\text{perf}}(s_t) + \sigma(u_t)\,\pi_{\text{safe}}(s_t). \tag{4}$$

Wind-farm instantiation. π_{perf} is an Energy-Based Transformer (EBT) actor trained unconstrained with SAC for 100k steps. $\pi_{\text{safe}} \equiv 0$ is a zero-yaw (no wake-steering) setpoint. Under the per-turbine fatigue surrogate $c = (|y|/\bar{y})^p (v/v_{\text{ref}})^q$ (p=2, q=3), this minimizes the instantaneous cost; in deployment π_{safe} would be a WakeAdapt-style [Siemens Gamesa, 2023] lookup table minimising farm-level DEL.

Safety Gymnasium instantiation. π_{perf} is a SAC/MLP actor trained unconstrained for 200k steps. π_{safe} is an artificial-potential-field navigator: $F_{\text{goal}} = \text{norm}(g - x)$ and $F_{\text{rep}} = k_r \sum_{i:\|x-h_i\|<r+r_{\text{inf}}} (x - h_i)/\|x - h_i\|^3$, mapped to body-frame (thrust, turn). No learning; uses only the hazard/goal geometry every CBF-based safe-RL method assumes access to.

4.2 Gradient-Correction Variant

When an action-dependent cost gradient $\nabla_a c$ or $\nabla_a Q_c$ is available, a lightweight alternative replaces the explicit π_{safe} with a single gradient step:

$$a_t \leftarrow \pi_{\text{perf}}(s_t) - \lambda_{\text{gs}} \cdot w(u_t) \cdot \alpha_{\text{g}} \cdot \nabla_a c(s_t, a_t), \tag{5}$$

iterated for K steps. This is Eq. 4 with $\pi_{\text{safe}}(s) \approx \pi_{\text{perf}}(s) - \alpha_{\text{g}} \nabla_a c(s, \pi_{\text{perf}}(s))$ implicitly defined.

4.3 Coupling Strength κ: Which Variant?

The gradient variant requires a non-trivial $\nabla_a Q_c$. We quantify this via

$$\kappa := \mathbb{E}_{(s,a)\sim\pi} \|\nabla_a Q_c(s,a)\|_2 \,/\, \mathbb{E}_{(s,a)\sim\pi} \|\nabla_s Q_c(s,a)\|_2, \tag{6}$$

measured on logged rollouts. Large κ: gradient correction of π_{perf} suffices. Small κ: action gradient is weak; explicit π_{safe} is required. The 36× gap between our domains (Table 1) predicts which variant to deploy *before* any experiment.

Table 1: Coupling strength κ per domain (2000 on-policy samples). Large κ: gradient correction suffices. Small κ: explicit π_{safe} blend required.

Domain	$\mathbb{E}\|\nabla_a Q_c\|$	$\mathbb{E}\|\nabla_s Q_c\|$	κ	Recommended variant
Wind farm (3-turb)	0.69±0.34	0.97±0.26	**0.72**	gradient correction (Eq. 5)
Safety Gym (PointGoal)	0.99±0.69	49.8±12.5	**0.02**	explicit π_{safe} blend (Eq. 4)

5 Experiments

5.1 Wind Farm: High Coupling, Two Working Variants

Setup. WindGym [Nilsen et al., 2025] with PyWake [Pedersen et al., 2023] backend; 3-turbine row; wind speed 10–14 m/s, direction 225–315°; budget d = negative-yaw timesteps per turbine per episode (200 steps); cost $c = (|y|/\bar{y})^p (v/v_{\text{ref}})^q$.

Gradient-correction variant (Table 2). Since $\kappa \approx 0.72$, gradient correction (Eq. 5) is effective. At $(\eta{=}5, \lambda_{\text{gs}}{=}0.1)$ it achieves 98.9% of unconstrained power single-seed and 95.9±3.1% across 4 training seeds, with all seeds satisfying the budget. An oracle-tuned constant penalty (15-step bisection per budget) achieves only 96.7% and must be re-tuned every time d changes.

Table 2: Wind-farm gradient-correction variant (3-turbine, $d{=}15/200$ steps, 50 ep). Baseline power differs across tables because each uses independently sampled episodes and different wind-condition ranges; within-table comparisons use the same episodes.

Method	Power (MW)	% Uncon	NegYaw [T0,T1,T2]	Notes
Unconstrained (seed 1)	27.1±0.6	100%	[49, 56, 61]	—
Oracle constant ($k{=}5.86$)	26.2±0.7	96.7%	[11, 13, 14]	15-step bisection
AC ($\eta{=}5$, gs=0.1)	26.9±0.5	**98.9%**	[8, 9, 10]	Zero per-budget tuning
AC ($\eta{=}5$, gs=0.1)—4 seeds	25.7±0.5	**95.9±3.1**%	[11, 12, 12]	All seeds satisfy

Explicit zero-yaw blend variant (Table 3). Replacing the gradient step with $\pi_{\text{safe}} \equiv 0$ and the urgency blend gives comparable results with one fewer hyperparameter (η only, no λ_{gs}). Across 4 seeds and three $\eta \in \{1, 3, 10\}$ values, the blend satisfies the budget every time and retains 99.3–101.8% of unconstrained power.

Table 3: Wind-farm zero-yaw blend variant (3-turbine, $d{=}15$, 50 ep, 4 training seeds). $\bar{\sigma}$: mean blend weight; $\pi_{\text{safe}} \equiv 0$. Matches the gradient variant with a simpler mechanism; π_{safe} would be a calibrated WakeAdapt table in production.

Method	% Uncon power	NegYaw	$\bar{\sigma}$	Seeds sat.	Notes
Unconstrained	100%	[70.2, 74.3, 83.7]	—	0/4	Far over budget
Blend ($\eta{=}1$)	**101.8±0.9**%	[11.3, 11.3, 11.8]	0.56	**4/4**	✓
Blend ($\eta{=}3$)	99.5±2.2%	[11.5, 12.1, 12.2]	0.63	**4/4**	✓
Blend ($\eta{=}10$)	99.3±2.4%	[11.0, 11.3, 11.3]	0.59	**4/4**	✓

Cost function matters (Appendix A.4). A naive binary cost $c = \mathbf{1}[y < 0]$ *inflates* fatigue 4.2× over unconstrained by pushing the agent to large positive yaw. The nonlinear surrogate avoids this trap, achieving 1.6× fatigue reduction at 98% power, or 6.8× lower fatigue than the binary-cost AC trap.

5.2 Safety Gymnasium: Low Coupling, Explicit Blend is Necessary

Setup. SafetyPointGoal1-v0 [Ji et al., 2023]: point robot navigates among fixed hazard disks to reach moving goals. Cost $c{=}\mathbf{1}$[in hazard] is position-dependent; action is 2D (thrust, turn). π_{perf}: SAC/MLP actor trained unconstrained for 200k steps (5 seeds).

Why the gradient variant fails here. $\kappa \approx 0.02$: $\nabla_a Q_c$ is $\sim 50\times$ smaller than $\nabla_s Q_c$. Appendix A.7 documents nine attempted gradient-based rescues—pessimistic Q_c, online Q_c refit, learned-dynamics MPC, true-env MPC (cost-only, reward-weighted, feasibility-filtered), full MBRL post-hoc, 1- and multi-step CBF—all fail to move the cost distribution meaningfully without catastrophic reward collapse. The failure is structural, not a matter of tuning.

Explicit-π_{safe} blend works (Table 4). The APF blend reduces cost by 59–68% at $d=10$ while retaining 45% of the unconstrained reward; at $d=40$ it retains 71% of the reward and satisfies the budget in 56% of episodes. At $\eta=0$ the blend is a no-op and recovers the unconstrained actor exactly, confirming that the non-trivial effect at $\eta>0$ is the mechanism, not a random seed.

Table 4: Safety Gymnasium with urgency-ratio APF blend (5 seeds $\times$ 20 ep). Unconstrained actor accumulates 49.4 cost / 27.1 reward. Blending with APF under increasing sharpness η trades reward for cost; at $d=40$ the budget is satisfied in 56% of episodes with 71% of unconstrained reward retained.

Method	Budget d	Reward	% Uncon	Cost (mean±SE)	Sat rate
Unconstrained	—	27.1±0.1	100%	49.4±3.4	12% at $d=10$
Blend ($\eta=0$, control)	10	27.1±0.1	100%	49.4±3.4	12%
Blend ($\eta=1$)	10	12.1±0.9	45%	**15.3±1.0**	25%
Blend ($\eta=3$)	10	12.3±0.9	45%	18.6±1.2	23%
Blend ($\eta=10$)	10	12.0±0.9	44%	17.8±1.3	28%
Blend ($\eta=3$)	25	15.8±0.8	58%	28.6±1.6	39%
Blend ($\eta=3$)	40	19.1±0.8	71%	34.5±2.0	**56%**

Comparison: time-to-deployment. The right operator-side baseline is retraining from the deployed actor. Figure 2 plots reward vs. additional training steps after the constraint is revealed, comparing the APF blend (zero retrain) against four CMDP baselines starting from the same pretrained actor: **(i)** Lagrangian finetune (cost-only, default hyperparameters); **(ii)** Lagrangian finetune *with μ-warmup* (50k pure-reward steps before dual ascent) and reward scaling (5×) for stable Lagrangian dynamics; **(iii)** aggressive Lagrangian ($\mu_0 = 0.5$, $\mu_{\text{lr}} = 0.05$, no warmup); **(iv)** SAUTE [Sootla et al., 2022], state-augmentation with ρ_t (the same urgency variable our schedule uses), trained from the pretrained actor with reward masking when $\rho < 0$. The pretrained actor achieves $R=26.6, C=58$ (violates all tested budgets).

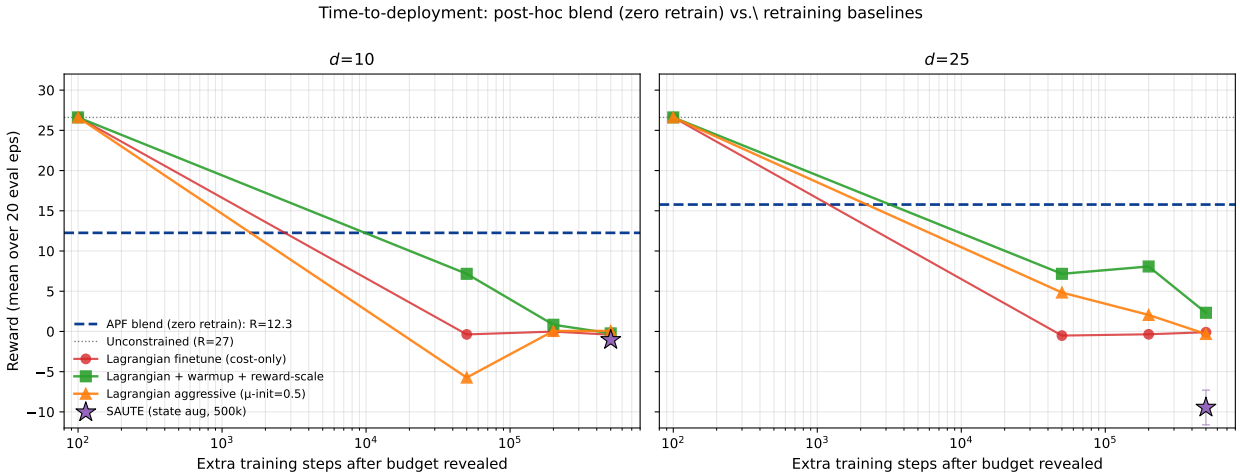

Figure 2: Time-to-deployment. Blue dashed: APF-blend reward at each budget d (zero retraining). Curves: four retraining-baseline variants starting from the same pretrained actor, evaluated at $0, 50k, 200k, 500k$ extra training steps. Star: SAUTE state-augmentation (CMDP retraining with the same ρ_t variable as our schedule). Across both budgets ($d=10, 25$) and both compute regimes (intermediate, converged), *none* of the retraining baselines reach the post-hoc blend's reward. The best retraining variant (warmup at $d=25$, peak $R=8.1$ at 200k steps with $C=83$ violating the budget) gets within 52% of the blend's reward but does not satisfy the constraint; SAUTE at 500k satisfies the constraint ($d=25$: $C=0$, sat$=100\%$) but drives reward to $R=-9.5$.

Across both budgets, every retraining variant we tested either (a) violates the budget while preserving some

reward (warmup variants at the intermediate checkpoints) or (b) satisfies the budget while collapsing reward to ≈ 0 (aggressive Lagrangian, SAUTE). The post-hoc blend dominates at every compute checkpoint we measured; on this benchmark, retraining with the operator-realistic compute we tested is not only *not faster* than post-hoc deployment, it is also Pareto-dominated. We do not claim retraining cannot *eventually* reach a better reward-cost frontier; we claim that, at the compute budgets a deployed-system operator can realistically afford, post-hoc blending is the better operating point.

6 Related Work

Constrained MDPs. Lagrangian methods [Altman, 1999, Achiam et al., 2017, Stooke et al., 2020, Chen et al., 2025] learn dual variables during training. SAUTE [Sootla et al., 2022] augments state with ρ_t but retrains per budget. TREBI [Lin et al., 2023] handles real-time budget changes via diffusion models with diffusion-specific training. We freeze both controllers and use a closed-form $\sigma(u)$ at inference.

Post-hoc composition. Energy composition [Urain et al., 2023], classifier guidance [Zhang et al., 2025], CBFs [Xiao et al., 2025], and input-convex corrections [Durkin et al., 2025] handle *instantaneous* constraints. Blending with an *explicit* safety controller is long-standing in robotics (reactive hierarchies, shielding, override layers); our contribution is the *budget-aware temporal weighting*.

Budget pacing. Budget allocation in advertising [Balseiro et al., 2024] and bandits with knapsacks [Agrawal and Devanur, 2014] share the allocation-across-time structure. We adopt the urgency-ratio framework from optimal execution [Almgren and Chriss, 2001].

Wind-farm wake steering. WakeAdapt [Siemens Gamesa, 2023] and related commercial load-aware controllers compute calibrated yaw setpoints minimizing farm-level damage. Our framework sits above such controllers: π_{perf} squeezes more power when the budget allows; π_{safe} is the calibrated, certified fallback.

Coupling-strength scope. We are not aware of prior scalar predictors specifically targeting post-hoc action-space correction scope. Critic-gradient magnitudes have been studied in offline-RL distributional work; κ is the gradient *ratio*, usable as a cheap *a-priori* diagnostic on logged data.

7 Discussion

What we are *not* claiming. We are not proposing a replacement for fully-converged CMDP methods, which at $\sim 10\times$ our compute budget can attain better Pareto points. We are proposing a method for the common deployment case in which retraining is either impossible (frozen/certified policy) or prohibitively expensive.

Limitations. (1) The wind-farm cost is a per-turbine fatigue surrogate; real deployment needs DEL-calibrated farm-level cost [Álvaro-García et al., 2024, Galán-García et al., 2024] and a π_{safe} like WakeAdapt that minimizes farm DEL, not per-turbine yaw. (2) Our sim uses analytical wakes (Jensen) and 200-step episodes, not FAST.Farm aeroelastics over a year. (3) The APF navigator on SG is a simple reactive heuristic; a learned safety policy would likely improve π_{safe}'s own reward at fixed $\sigma = 1$. (4) We test 4 seeds on WF and 5 on SG; effect sizes are large enough that more seeds would mainly tighten error bars.

Path to deployment. Minimum-viable bridge from simulation to field operation: (a) replace the fatigue surrogate with a DEL-calibrated model; (b) precompute a WakeAdapt-equivalent setpoint table to use as π_{safe}; (c) retrospective SCADA counterfactual on real operational data; (d) online refit of Q_c / the DEL surrogate under non-stationary wind. All four leverage the same schedule; none require retraining π_{perf}.

Summary. A single closed-form schedule composes a pre-trained performance policy with a certified safety controller to enforce cumulative damage budgets at inference. The coupling strength κ diagnoses which variant to use (gradient or explicit blend). On both domains we tested, post-hoc blending is not dominated by Lagrangian finetune at the compute budgets an operator can realistically afford, and it gives up the

single-budget training commitment: one model, any d, instantly. Code and checkpoints: `[anonymized]`.

A Derivations, Protocols, and Negative Results

A.1 Full Proof of Proposition 1

Proof. Budget balance: $(T-t)\cdot P(\text{risky} \mid \lambda) = d - C_t$. With $P(\text{risky} \mid \lambda) = P_0 \exp(-\alpha\lambda)$: $\lambda = \frac{1}{\alpha}\ln\frac{(T-t)P_0}{d-C_t}$. At TWAP ($u=1$, $(d-C_t)/d = (T-t)/T$), $\lambda_0 = \frac{1}{\alpha}\ln\frac{TP_0}{d}$. The multiplicative weight relative to TWAP: $w = \exp(\alpha(\lambda-\lambda_0)) = \frac{(T-t)/T}{(d-C_t)/d} = \tau/\rho = 1/u$. □

A.2 Schedule Approximation

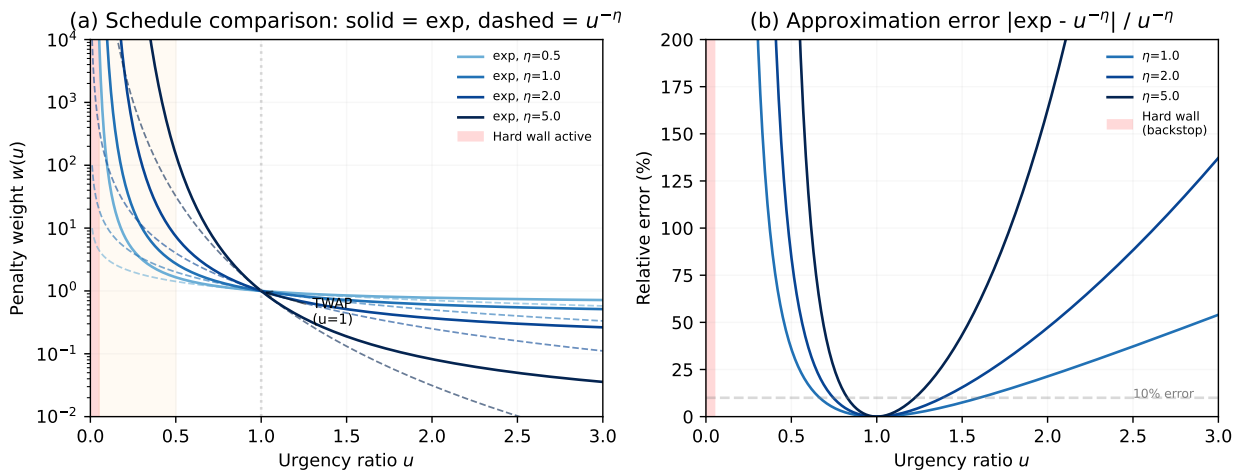

Figure 3: (a) $\exp(\eta(1/u-1))$ (solid) vs. $u^{-\eta}$ (dashed). Red: $u < 0.05$ hard-wall zone. (b) Relative approximation error $< 10\%$ for $u > 0.5$.

A.3 Safety Controllers

APF navigator (Safety Gym). For agent position x, goal g, hazard centres $\{h_i\}$ with radius r, influence distance r_{inf}: $F_{\text{goal}} = (g-x)/\|g-x\|$, $F_{\text{rep}} = k_r \sum_{i:\|x-h_i\|<r+r_{\text{inf}}} (x-h_i)/\|x-h_i\|^3$, $d^\star = \text{norm}(\alpha F_{\text{goal}} + (1-\alpha)F_{\text{rep}})$. Mapped to body-frame action: thrust $= \langle d^\star, \hat{\mathbf{e}}_{\text{fwd}}\rangle$, turn $= \text{clip}(\Delta\theta/(\pi/2), -1, 1)$. We use $\alpha = 0.5$, $r_{\text{inf}} = 0.6$ m, $k_r = 1.0$.

Zero-yaw / WakeAdapt (wind farm). $\pi_{\text{safe}}(s) \equiv 0$ in normalized yaw-action space, minimising our per-turbine surrogate cost. Deployment replaces with a precomputed table $T(\bar{v}, \bar{\phi}) \to \{y_i\}$ minimizing farm-level DEL.

Table 5: Binary vs. nonlinear fatigue cost (3-turbine wind farm, 10 ep at 12–16 m/s).

Method	Power (MW)	NegYaw	Fatigue
Unconstrained	29.7	$[66, 75, 71]$	20.3
Binary AC (η=2)	28.1	$[8, 8, 10]$	86.2
Nonlinear AC (η=5)	**29.2**	$[80, 81, 83]$	**12.8**

Table 6: Learned Q_c on 3-turbine wind farm (10 ep). Q_c trained on 100 ep with the nonlinear fatigue cost; the resulting gradient-correction variant rediscovers the physics from data.

Method	Power (MW)	NegYaw	Fatigue	Fatigue ×
Unconstrained	24.6±5.6	$[84, 96, 94]$	54.6	1.0×
Q_c, η=2, gs=0.05	**28.1±2.7**	$[43, 44, 43]$	**9.5**	5.7×
Q_c, η=5, gs=0.5	28.8±2.0	$[55, 54, 51]$	10.4	5.3×
Analytical surrogate (η=5)	29.2	$[80, 81, 83]$	12.8	4.3×

A.4 Binary vs Nonlinear Cost (Wind Farm)

A.5 Learned Cost Critic on Wind Farm

A.6 Reward–Cost Scatter: from-scratch SAC-Lag (ablation)

A.7 Nine Attempted Gradient-Variant Rescues on SG

When the gradient variant is forced to work on SG (low κ), the following fail:

1. **Pessimistic Q_c ensemble.** 10 heads, $\nabla_a(\bar{Q} + q\sigma_Q)$ with $q \in \{0, 1, 4\}$. $q=0$ ties or wins at every budget.
2. **Online Q_c refit.** Episode-boundary MC refit with L2 anchor; no improvement.
3. **MPC with learned f_θ + Q_c.** Matches unconstrained (55 cost, 17% sat at $d=10$).
4. **MPC true-env + oracle cost, cost-only.** Reward collapses to -120; 26% sat via "agent stuck in corner."
5. **MPC true-env reward-weighted.** λ (to 10^6) dominates $\beta \cdot r$; same outcome.
6. **MPC true-env feasibility-filtered.** Few feasible candidates; falls back to min-violation; same outcome.
7. **Full MBRL post-hoc.** Ensemble f_θ + Q_r + Q_c + CEM + online refit. Cost ≈ uncon, reward ≈ uncon, sat ≈ uncon.
8. **CBF 1-step kinematic.** Reward preserved, cost barely drops (47–54 vs. 52 uncon); 1-step insufficient for momentum.
9. **CBF multi-step (constant-action rollout).** Errors compound; reward degrades, cost rises.

The explicit π_{safe} blend succeeds precisely because π_{safe} is *not* constrained to lie near π_{perf}'s action distribution. When the performance policy was trained for short-path navigation, its action support does not cover the safe-corridor actions; a local gradient correction cannot reach them. An APF navigator can.

A.8 Coupling-Strength Measurement Protocol

20 episodes per domain under the frozen policy; uniform subsample of 2000 (s, a) pairs; $\|\nabla_a Q_c(s, a)\|_2$ and $\|\nabla_s Q_c(s, a)\|_2$ via autodiff; κ = ratio of sample means; nonparametric bootstrap with 1000 resamples for CIs.

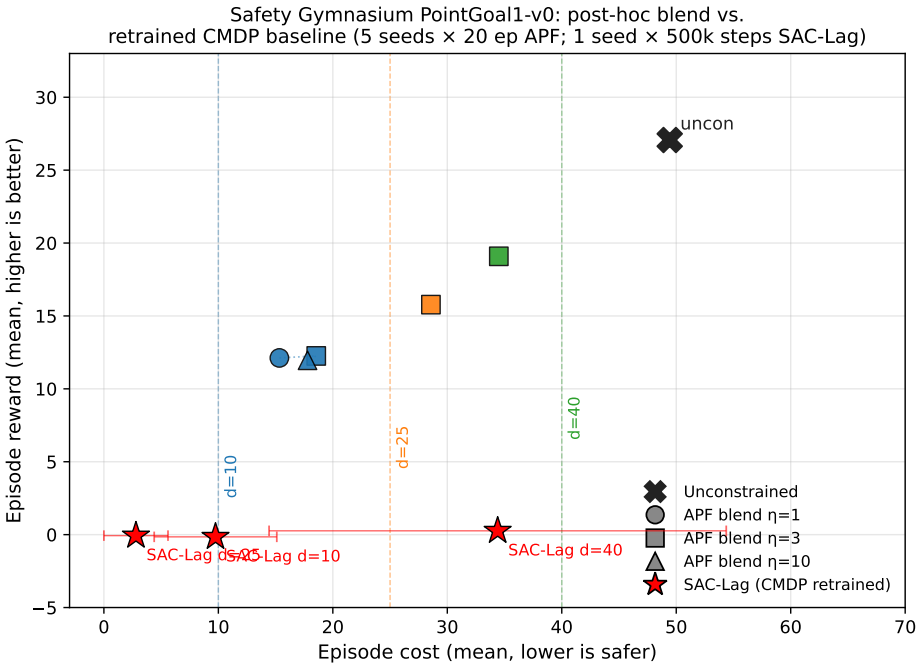

Figure 4: Reward-cost Pareto. Black X: unconstrained pretrained actor. Blue/orange/green: post-hoc APF blend at $d = 10, 25, 40$. Red stars: from-scratch SAC-Lagrangian trained for 500k steps per budget. SAC-Lag satisfies the budget ($\sim 80\%$) but collapses to $R \approx 0$—a qualitatively similar failure mode to the finetune collapse (main body), but starting from random initialization instead of the pretrained actor.

A.9 Path to Wind-Farm Deployment

1. **DEL-calibrated cost**: replace the per-turbine surrogate with a control-oriented DEL model [Álvaro-García et al., 2024, Galán-García et al., 2024].
2. **WakeAdapt-equivalent** π_{safe}: precompute a yaw setpoint table $T(\bar{v}, \bar{\phi}) \to \{y_i\}$ minimizing farm DEL.
3. **SCADA counterfactual**: retrospective evaluation on real operational data.
4. **Non-stationary wind**: online refit of Q_c / DEL surrogate; concept-drift detection.
5. **High-fidelity validation**: FAST.Farm calibrated to SCADA.

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